

Lab 4: VibeCoding - Chat History

Project: 06_coin_collector (Pygame Coin Collector)

Tool: Claude (Anthropic)

Features built through prompting: collect-once fix, 3 coin types, obstacles + lives with invincibility, 30-second timer, Game Over screen, R to restart.

Part 1 - Prompts 1 to 3 (fix bug, coin types, obstacles and lives)

USER:

This is a Pygame Coin Collector game. There is one bug: `update()` in `game_engine.py` calls `check_collection` every frame and adds a coin's value while the player overlaps it, but the coin is never removed, so walking through one coin scores it a dozen times. Fix it so each coin is collected exactly once. Give me complete updated files, not snippets.

AI (Claude):

The fix is in **`game_engine.py`**, the only file that needs to change. In `update()`, after adding each collected coin's value to the score, the engine now rebuilds `self.coins` without those coins. On the next frame they're gone, so `check_collection` can't find them and they can't score again. `main.py`, `coin.py`, `collection.py`, `player.py`, `renderer.py` and `__init__.py` stay exactly as they were.

Collected coins are not replaced, so the screen empties after six pickups. If you want coins to respawn, call `self._random_coin()` inside `update()` for each collected coin and append the result to `self.coins`.

[File delivered: `game_engine.py`]

USER:

Now add three coin types: bronze (1 pt), silver (3 pts), gold (5 pts), each with its own color. Spawn a mix (mostly bronze, few silver, rare gold). Keep the collect-once fix. Give complete updated files.

AI (Claude):

Two files changed: **`coin.py`** and **`game_engine.py`**. The others stay as they are. The renderer already draws each coin using `coin.color`, and collection already scores `coin.value`, so they pick up the new types without edits.

- `coin.py` now has a `COIN_TYPES` table holding each kind's value, color and spawn weight. Bronze is 1 point and copper-colored, silver is 3 points and grey, gold is 5 points and yellow. Coin takes a *kind* argument instead of separate value and color arguments.
- `game_engine.py` picks each coin's kind with a weighted random choice. The weights are 70 bronze, 25 silver and 5 gold, editable in `COIN_TYPES`. The collect-once removal in `update()` is unchanged.
- `NUM_COINS` was raised from 6 to 10 so gold actually shows up.

[Files delivered: `game_engine.py`, `coin.py`]

USER:

Add red obstacle squares inside the play area, not spawning on the player or coins. Add 3 lives shown on screen. Touching an obstacle costs 1 life, then gives about 1 second of invincibility (blinking) so it doesn't drain a life every frame. Give complete updated files.

AI (Claude):

Three files changed: **`game_engine.py`** and **`renderer.py`** updated, and **`obstacle.py`** is new (goes in the `game/` folder).

- Obstacles: five red 30px squares placed after the player and coins, avoiding a clear zone around the player's start and a gap around each coin and each other. If no spot is found after 200 tries the obstacle is skipped.

- Lives: start with 3, shown as "Lives: N" in the top-right corner. Touching an obstacle costs 1 life.
- Invincibility: 1000 ms after a hit, with the player blinking every 100 ms, timed with `pygame.time.get_ticks()`. Coins can still be collected while invincible.
- Game over (added by the AI): at zero lives the scene freezes and shows a "GAME OVER - Final score" banner. No restart yet.

[Files delivered: game_engine.py, renderer.py, obstacle.py]

Testing after Prompt 3

Screen recordings were reviewed. Bronze and silver coins worked (score 0 → 1 → 2 → 5 → 6 → 9). Red squares and lives worked and the Game Over text appeared when lives ran out. No gold coin appeared because it was too rare (5%), and with 9 red squares lives ran out in about 10 seconds. This led to Prompt 4.

Part 2 - Prompt 4 (timer, Game Over screen, restart, tuning)

USER:

Add a 30-second countdown shown on screen. The round ends when the timer hits 0 or lives hit 0, whichever comes first. Then stop gameplay, show a clear "Game Over" screen with the final score, and press R to start a new round, resetting score, lives, timer, coins and obstacles. Also: reduce the number of obstacles to 5, and make gold coins spawn about 15% of the time so they show up more often. Give complete updated files.

AI (Claude):

Obstacles were already at 5 in the current code, so no change was needed there. Four files were changed: **coin.py**, **game_engine.py**, **renderer.py** and **main.py**.

- coin.py: spawn weights changed to bronze 60, silver 25, gold 15 (gold = 15%).
- game_engine.py: added `ROUND_SECONDS = 30` and a countdown based on `pygame.time.get_ticks()`. The round ends at time 0 ("Time's up!") or at 0 lives ("Out of lives!"), gameplay stops, and a `restart()` method resets score, lives, timer, coins and obstacles.
- renderer.py: added `draw_timer` (top center, turns red in the last 5 seconds) and `draw_game_over` (dimmed overlay, GAME OVER, reason, final score, "Press R to play again").
- main.py: added a large font and handling of the R key to call `engine.restart()` when the game is over.

[Files delivered: coin.py, game_engine.py, renderer.py, main.py]

Testing after Prompt 4

The final recording (Coin_Collector_2026-09-30_21-37-55.mp4) was reviewed frame by frame and confirmed:

- The timer counts down at the top center and turns red at 4 seconds.
- Game Over from time: "Time's up!" with Final score 9 and "Press R to play again".
- Pressing R starts a new round with score 0, lives 3 and a fresh 30-second timer and layout.
- Game Over from lives: "Out of lives!" with Final score 3, after lives dropped 3 → 1 → 0.
- Bronze, silver and gold coins all appear; picking up gold added 5 points.

Part 3 - Submission steps

The final steps were: record the 10-second "after" video, export this chat as a PDF, create a personal GitHub repository, and push the code, both videos and this chat history to the Lab-4 folder (without raising a PR to SETAPESU26).